

Computational supplement for A backreacting phase clock with fixed momentum in pure-Weyl gravity

GPT-5.6.sol*

Theorem-frozen version, July 2026

Abstract

This supplement records the claim-to-certificate map, sparse-operation inventory, replay commands and release gates for `09-relational-clocks-berger-d-cartan.tex`. It separates mathematical arguments from machine receipts and keeps the paper fail-closed: a certificate supports only the claim inside its declared phase space and boundary. The manuscript is marked `THEOREM_FROZEN` for the ten classical gravity-clock claims below. Maxwell and observer results are not main-theorem dependencies; a separate publication-current ledger now types their conditional, carrier-scoped and nonactivation dispositions.

The authorship line describes the research archive. Venue-specific human accountability and authorship-policy review remain release questions and are not silently treated as completed by this supplement.

1 Focused claim ledger

The authoritative machine-readable ledger is `../d_quotient_classical/certificates/PAPER_09_BERGER_CLAIM_TABLE.json`. The table below is its human-readable projection. The publication-current superset, including the observer dispositions, is `09-relational-clocks-berger-d-cartan-claim-map.json`; it imports this ten-claim ledger without replacing it.

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

Claim	Mathematical content	Authoritative certificate
P09-C1	Exact smooth non-conformally-flat Berger clock family	POSITIVE_BERGER_CLOCK_BACKGROUND.json
P09-C2	Standard-sign target, timelike phase, positive quartic and energy inequalities	POSITIVE_BERGER_CLOCK_BACKGROUND.json
P09-C3	Internal current, nonzero Q_R and $\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega\delta Q_R$	BERGER_CLOCK_REDUCED_CHARGE_SEED.json
P09-C4	Fixed-coupling lapse constraint, compact averaging, $\delta Q_R = 0$ and linear presymplectic nullity	BERGER_FIXED_COUPLING_DELTA_CHARGE.json
P09-C5	Complete cyclic gauge-fixed 54-row unary complex and support-local $54 \rightarrow 26$ contraction	BERGER_GAUGE_FIXED_NONMINIMAL_COMPLETION.json
P09-C6	Advanced/retarded all-row chain contractions, support, adjointness and K -equivariance	BERGER_54_ROW_CAUSAL_GREEN_HOMOTOPY_V2.json
P09-C7	Complete arbitrary-input support-local q_2 , cyclicity, L_∞ identity and K derivation	BERGER_SUPPORT_LOCAL_Q2.json
P09-C8	Complete arbitrary-input support-local q_3 , quartic cyclicity, arity-three identity and $L_K^{(3)} = 0$	BERGER_SUPPORT_LOCAL_Q3.json
P09-C9	Cyclic causal K -Cartan contraction through arity two	BERGER_CAUSAL_D_CARTAN_V2.json
P09-C10	Closed arity-three K -source and cyclic C_4 causal primitive on all 54 rows	BERGER_ARITY_THREE_D_CARTAN_FULL_4D.json

All paths in the JSON ledger are relative to `d_quotient_classical/certificates/` and are pinned by SHA-256. The verifier rejects missing files, digest drift, a false required flag, a true forbidden downstream flag, or a manuscript claim identifier absent from both paper sources.

Two independent interpretation checks are pinned separately. The exact co-rotating audit `BERGER_GENERATOR_CONJUGATION_AUDIT.json` proves that the legacy all-row e_0 operation represents $K = D - \omega R$, while raw D has a nonzero zeroth Taylor component. It supports P09-C6–P09-C10 and forbids interpreting those certificates as an affine D -Cartan theorem. The strategic producer-boundary check is `BERGER_Q3_ACTION_SECTOR_CROSSCHECK.json`; it supports the action-to-payload provenance of P09-C8 in two exact sectors but does not replace the complete `BERGER_SUPPORT_LOCAL_Q3.json` certificate.

2 Observer publication disposition

The observer column is deliberately fail-closed. The following entries are material Paper 9 statements, but none is promoted into the ten classical main theorems.

Claim	Type	Publication disposition
P09-01	CONDITIONAL_THEOREM	Charged-time event current; nonzero evaluation still requires a descended receiver period.
P09-02	CONDITIONAL_THEOREM	Finite-resolution sampling and its profile-error bound preserve that condition.
P09-03	CONDITIONAL_THEOREM	Reciprocity, composition and loop laws require retarded paths and explicit fibre crosswalks.
P09-04	CONDITIONAL_THEOREM	Receiver admissibility is a certified interface, not a populated receiver.
P09-05	KINEMATIC_FIXTURE	The legacy $G0$ value $1 + z = 2$ survives only as a spatially global retarded probe-mode result; it is not operational redshift.
P09-06	KINEMATIC_FIXTURE	Three distinct pre-quotient source-to-probe maps retain exact rank two on their own carriers.
P09-07	NOT_ACTIVATED	The legacy frequency-ratio domain is absent; its coordinate ratio one is not relational redshift.
P09-08	LOCAL_BV_CLASS	The standalone $D0$ memory-shift receiver has a local BV descent only.
P09-09	ACTION_INTEGRATION	A degree-zero local-cochain-to-repaired- q_{70} -chain map is exactly obstructed.
P09-010	PHYSICAL_DESCENT	Phase 1 ends with no residual nonradical receiver period and no operational frequency-ratio map.
P09-011	OPEN	For $j \geq \frac{3}{2}$ the quotient, divisor and pairing have <code>NO_CERTIFIED_MAP</code> .
P09-012	NOT_ACTIVATED	All certified $j = 0, \frac{1}{2}, 1$ repaired- q_{70} blocks are unstable; fixed Q_{rel} removes the clock, so the thirteen-field operational ratio domain is empty.

The machine ledger consequently records zero entries of type `OPERATIONAL_OBSERVABLE`. This zero is a checked disposition, not an omission. Each imported result, payload, receipt, terminal lifecycle event and transitive health input has an exact hash, a materiality annotation and a result-to-paper edge. The independent verifier rejects a stale healthy carrier, a coordinate ratio relabelled as redshift, or an identification $K = D$.

3 Invariant complex and component realization

The invariant retained complex consists of spatial diffeomorphism ghosts, the dressed symmetric metric, its antifield, and the spatial identity row, with component ranks

$$3 \longrightarrow 10 \longrightarrow 10 \longrightarrow 3.$$

The 54-component gauge-fixed realization adds eight minimal clock, temporal-diffeomorphism and Weyl components and twenty nonminimal components:

$$54 = 26 + 8 + 20 = 26 + 28.$$

The exact degree decomposition is

$$5z^{-1} + 22 + 22z + 5z^2.$$

The retained 26 components are the propagating/constraint endpoint of the support-local contraction; the complementary 28 components form explicit contractible pairs. This separation keeps the theorem invariant under a different ordering of the same gauge-fixed rows.

4 Exact operation inventory

Operation	Rows	Nonzero rows	Sparse PBW terms	Maximum jet order
q_2	54	39	150,305	6
q_3	54	12	5,812,130	8

Both operations use exact coefficients in $\mathbb{Q}(\sqrt{10})$ at the rational Berger fixture. The q_3 portable representation is a strict-JSON manifest plus 54 deterministic gzip-compressed strict-JSON row chunks. Compression is storage only; the operation acts on arbitrary four-dimensional jets and is not a reduced-mode artifact.

The Taylor normalization is

$$Q(\phi) = q_1(\phi) + \frac{1}{2}q_2(\phi, \phi) + \frac{1}{6}q_3(\phi, \phi, \phi) + O(\phi^4),$$

and the invariant producer contract is

$$\langle x_0, q_n(x_1, \dots, x_n) \rangle_{\text{BV}} = \partial_{t_0} \cdots \partial_{t_n} S_{\text{BV}}[\bar{\Phi} + \sum t_i x_i] \Big|_{t=0}.$$

The PBW payload is therefore an implementation of a variationally defined operation, not its definition.

The recorded q_3 Tier-2 stages are:

1. action differentiation on the raw minimal complex;
2. canonical clock and gauge-fermion transport to all 54 rows;
3. row-bounded exact replay of graded symmetry, Q^2 , quartic cyclicity and K derivation (under the generator-conjugation audit);
4. portable row-chunk finalization;
5. independent schema, hash, PBW and adjacent-Koszul audit.

The independent consumer audits the frozen operation but does not independently rederive the Bach-clock expansion from the action. The paper states this distinction explicitly.

5 Independent action-to-q3 sector check

The targeted second route writes the homogeneous reduced density independently in

$$h = \hat{h}_{00}, \quad r = R, qquad N = \sqrt{1 - h - 2r}, \quad k = 1 - 2r, qquad \rho = 1 + r,$$

as

$$\mathcal{L}(h, r) = Nk^{3/2} \left[\frac{\alpha_B \bar{C}^2}{8k^2} + \frac{\rho^2 \omega^2}{2N^2} - \frac{\bar{R} \rho^2}{12k} - \frac{\lambda \rho^4}{4} \right].$$

It imports neither the jet classes nor the q_3 producer. In the frozen twice-Euler-density convention, the direct fourth derivatives give

Output	$h^0 r^3$	$h^1 r^2$	$h^2 r^1$	$h^3 r^0$
$\hat{h}_{00}^*$	-27	27/8	81/16	405/128
R^*	-1071/40	-27	27/8	81/16

All sixteen ordered permutations agree with the published row chunks. This is a genuine action-to-payload cross-check in the lapse and dressed radial/Weyl directions. It does not independently regenerate the remaining millions of coefficients, and the certificate forbids that broader reading.

6 Generator-conjugation audit

With $R(T_1, T_2) = (-T_2, T_1)$ and $U(t) = \exp(\omega t R)$, write $T = U(t)((\rho, 0) + \psi)$. Direct exact differentiation gives

$$U^{-1}DT = \partial_t \psi + \omega R \psi + \omega R(\rho, 0), \quad U^{-1}(D - \omega R)T = \partial_t \psi.$$

The frozen 54-row coordinate rule is $e_0 I_{54}$, hence it represents $K = D - \omega R$. The strict audit pins both the frozen unary certificate and the charge convention, and requires

Flag	Value
EXPORTED_UNARY_GENERATOR_IS_K	true
EXPORTED_UNARY_GENERATOR_IS_ORIGINAL_D	false
AFFINE_D_ZERO_ARITY_NONZERO	true
AFFINE_D_CARTAN_CONSTRUCTED	false

This corrects the interpretation of the legacy artifact names without changing their coefficientwise algebra.

7 Causal domains

The unary maps have the typed domains

$$\Lambda_{54, \pm} : \Gamma_c(\mathcal{C}_{54}) \longrightarrow \Gamma_{J^\pm}(\mathcal{C}_{54})[-1].$$

The local Taylor operations preserve intersections of input supports. The cyclic higher primitives use both time orientations and land in the two-sided causal hull. They are not the causal difference $\Lambda_+ - \Lambda_-$ and are not claimed to be separately retarded or advanced.

8 Cartan sign and support audit

At arity three, the source

$$A_K^{(3)} = [q_3, \iota_K^{(1)}] + [q_2, \iota_K^{(2)}] - L_K^{(3)}$$

has two normalized closure channels. Their exact coefficients are

$$-\frac{1}{2} + \frac{1}{2} = 0, \quad [q_3, K] = 0.$$

The first equality is graded Jacobi; the second is the imported local K -derivation identity with $L_K^{(3)} = 0$.

For cyclicization, the operator output is first paired with the degree-one odd Darboux form. The resulting four-linear tensor uses the Koszul rotation

$$(-1)^{|x_0|(|x_1|+|x_2|+|x_3|)}.$$

The actual degree multiplicities are

$$5z^{-1} + 22 + 22z + 5z^2.$$

The coefficient of z^0 in its fourth power is 978,736. Exhaustion over these actual degree classes gives zero C_4 group-law defects. Exact group algebra arithmetic verifies

$$\left(\frac{1 + \tau + \tau^2 + \tau^3}{4}\right)^2 = \frac{1 + \tau + \tau^2 + \tau^3}{4}.$$

The support claim is

$$\text{supp } \iota_{K, \text{cyc}}^{(3)}(f, g, h) \subseteq J(\text{supp } f \cup \text{supp } g \cup \text{supp } h).$$

The certificate deliberately rejects the stronger statement that this cyclic primitive is separately retarded or advanced.

9 Fast replay and frozen heavy tier

From the repository root for this programme, the focused fast checks are:

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_causal_d_cartan_arity_three.py
```

```
PYTHONPATH=. pytest -q \
  d_quotient_classical/backreacted_clock/tests/\
  test_berger_causal_d_cartan_arity_three.py \
  d_quotient_classical/backreacted_clock/tests/\
  test_paper_09_claim_table.py
```

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_q3_action_sector_crosscheck.py --check --guards
```

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_q3_action_sector_crosscheck.py
```

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_generator_conjugation_audit.py --check --guards
```

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_generator_conjugation_audit.py
```

The complete q_3 derivation is not part of this fast rail. Its recorded Tier-2 receipt is `../d_quotient_classical/certificates/BERGER_SUPPORT_LOCAL_Q3_TIER2_RECEIPT.json`. Recompute that tier only when the action, convention, canonical maps, PBW engine or payload changes; otherwise verify and import its content-addressed output.

10 Fail-closed boundaries

The main theorem depends on the conjunction of the ten claims above. It does not promote any of the following:

- a curved/affine D -Cartan contraction at any nonlinear order;
- an unconditional or convergent all-orders K -Cartan contraction;
- arity-four closure;
- an integrated nonlinear gauge quotient or global complete observable;
- a general-background nonlinear Koszul–Tate export;
- Hadamard or Feynman two-point functions;
- renormalized composites, anomaly cancellation or the quantum master equation;
- a positive graviton Hilbert space or interacting unitarity;
- a boundary, asymptotic, black-hole or scattering charge theorem.

The positive matter clock and the vanishing compact-universe tangent charge must not be transported to those settings without a new phase-space and boundary calculation.

11 Post-freeze two-phase counterflow successor disposition

A separate two-phase counterflow successor does not furnish a healthier replacement for the frozen theorem. Its repaired 70-row causal BV parent is now physically quotiented on the maximal certified Peter–Weyl domain

$$j = 0, \quad j = \frac{1}{2}, \quad j = 1,$$

with every left and right weight retained. Across the full degeneracy orbits these blocks contain 980 q70 coordinates, 364 coordinates after the 26-row contraction, and 98 nonzero-frequency physical quotient coordinates. They contain, respectively, a real exponential direction, a Hamiltonian–Hopf quartet, and complex-frequency sectors; every displayed characteristic pairing has zero radical.

All nonzero-frequency witnesses survive the fixed- Q_{rel} restriction. Fixing the charge instead removes the separate global relative action–angle pair. Linear physical health is therefore obstructed on both charge branches. This conclusion is complete as an obstruction but not as a mode census: for $j \geq \frac{3}{2}$ the support-local causal parent is imported while the physical quotient, characteristic divisor and pairing inertia are all NO_CERTIFIED_MAP. No higher- j spectrum is extrapolated from the first generic counterexample. The certificate is `../d_quotient_classical/compensator/TW0_PHASE_COUNTERFLOW_REPAIRED_Q70_HEALTH_ASSEMBLY_MAXIMAL_DOMAIN_V1.json`.

12 Theorem-freeze receipt

The frozen release satisfied the following checklist:

1. every pinned certificate and independent verifier passes from a clean committed tree;

2. the release manifest records the source commit, operating system, Python, SymPy, JSON Schema and pytest versions, and distinguishes the fast and heavy rails;
3. the JSON claim ledger, this table and the manuscript theorem have no missing or broadened dependency;
4. the nonlinear team independently imports the frozen q_2 , q_3 and arity-three Cartan hashes and signs off Sections 6–7;
5. the action-to- q_3 strategic cross-check remains exact and is described as partial rather than a full second producer;
6. the generator-conjugation audit remains exact, the BV theorem is stated for K , and no legacy artifact name is read as an affine D theorem;
7. the abstract, theorem and conclusion retain the classical, fixed-coupling, compact-Berger and arity-three limitations;
8. the compiled main paper and supplement are reproduced from a clean checkout.

The original machine-readable replay and environment manifest is `../d_quotient_classical/certificates/PAPER_09_THEOREM_FREEZE.json`. The publication-current observer audit is `09-relational-clocks-berger-d-cartan-claim-map.json`, independently checked by `verify_09_relational_clocks_claim_map.py`. The original ten scientific claims retain their frozen scope, but this publication-current edition is **DRAFT_ALLOWED**: the legacy ledger’s TeX source binding must be formally superseded or repinned, and the pre-existing 84-row apparatus certificate drift must be repaired before an uninterrupted full observer suite can pass. This disposition records the observer nonactivation; it does not promote Maxwell apparatus, Hadamard, QME or all-orders claims.